

What's New In Wiring 1.0 Build 101

Wiring 1.0 supports almost 80% of the Atmel megaAVR microcontrollers with its Atmel 8-bit AVR Framework core.

Improvements

- Add support for OS X 10.9 and 10.10
- Replace rxtx library
- Add support for Windows 8.1
- Fixed random and map issue with longs

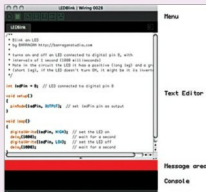


Fig. 2: Screen Wiring

small programs and downloading to the input/output board.

Wiring input/output board

Let us take a look first at the Wiring I/O board. It is a simple electronic system that includes a microcontroller where the users' programs run. By working with digital and analogue inputs and outputs, it combines form and functionality to let its users interface all kind of sensors and actuators with the board.

The board also allows you to program the behaviours needed in your projects. Rather than focusing on showing that the technology works, users are encouraged to utilise their time to work on the quality of design, the interaction effect, ideas and concept illustration.

The active user community contributes heavily to the wider Interaction Design community. The libraries have been separated into categories. While the core libraries are coded for specific hardware (microcontrollers), cross-platform



Fig. 3: Span of Wiring

libraries are those that do not depend on hardware. And of course, you can forget the contributed libraries that come from all over the Internet.

Interfacing with sensors and actuators

The numerous libraries that support Wiring enable it to go beyond the basic input/output features. These allow it to interface with sensors, actuators and other devices.

EEPROM library. It helps in easily reading and storing data with the board's internal electrically erasable programmable read only memory (EEPROM). It is ideal to store data that does not change often—like permanent data strings or application configuration details.

LiquidCrystal library. It allows for easily writing data to a liquid crystal display (LCD), thus allowing an output for the program running on Wiring.

SoftwareSerial library. The Wiring SoftwareSerial library let's Wiring read and write data to and from external devices through software serial ports. It allows two devices to communicate with each other.

ServoMotor library. Lets you man-

ipulate a standard servo motor (up to 24 motors) connected to a pin. This is helpful for projects relating to airplane control, robotics and precision tasks.

NewSoftSerial library. This is a new software serial library that provides interrupt-driven receives instead of the polling required by the SoftwareSerial library mentioned earlier.

Wiring hardware can also be interfaced with other tools including Blender, Pure Data, MaxMSP, Adobe Flash and more. Blender offers a free and open source 3D creation suit that supports a lot of video editing and gaming features. You can set up a Wiring board to play games on Blender. Pure Data, Cycling and Isadora also allow many similar creative possibilities.

You don't need Wiring board to use Wiring

Yes, you don't really need a Wiring board to use Wiring. If you are an advanced user, look under the Tools menu where you will find a selection of manufacturers. The definitions can be found in the Tools→Board menu for popular bare ATMEGA DIP parts like atmega168p, ATmega328p, ATmega168 and ATmega644p. Wiring also supports directly the Arduino hardware, Rogue Robotics, BDMicro and many others.

Hernando Barragán, Brett Hagman, Alexander Brevig and their team manage the open source Wiring software. The PDE (processing development environment) is released under the GNU GPL (general public licence). The export libraries (also known as 'core') are released under the GNU LGPL (lesser general public licence).

Interestingly enough, the name Wiring originated as the action of 'connecting things one to another' and also from the Processing project. A pretty nice thought, if you think about it. **EFY**

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